# openQSE Community

A short presentation on how the different members and communities can work together to realize an openQSE specification

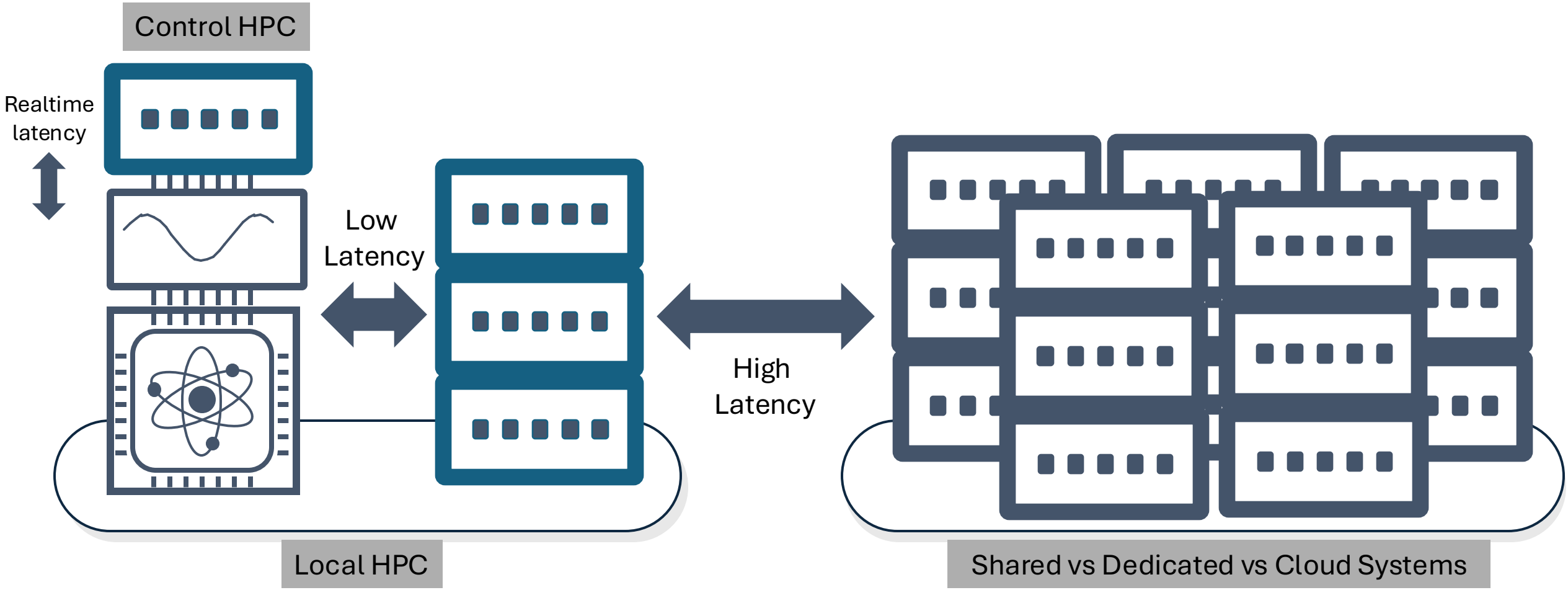
# Welcome to the openQSE community!

- This is a public forum for developers, end users, and stakeholders to create software specifications for quantum-HPC systems.
- Our goal is to define an **open QHPC software ecosystem (openQSE)** that provides guidance for HPC data centers and quantum computer vendors.
- Our approach is to use technical exchanges, voluntary contributions, and agile development to create well-documented specifications and, when possible, reference implementations.

# Why do this now?

- We are witnessing major advances in the capabilities and capacities of quantum computing systems.
- Efficient integration of quantum computer systems with HPC resources will require coordination.
- We believe that a community effort can address the critical concern of ensuring portability and interoperability across platforms and data centers.

# What is QHPC integration?



# How are we developing openQSE?

- We are very early in the development process to establish an open, community-driven specification
- Our first step has been to recruit members that can define the scope of an effort and ensure clear coordination.
- We recruited many key contributors, and we are in the process of formalizing the scope and structure.
- Next steps will finalize governance and change control processes.
- We need to develop technical priorities to ensure progress is being made.

# We expect several public repositories of information (names are TBD)

- **openQSE-spec** is a public specification of the major components and interfaces expected for QHPC integration.
- **openQSE-ref** is a public reference implantation of the specification.
- **openQSE-test** is a public test suite to confirm adherence to the specification.
- **openQSE-impl** are your independent implementations of the specification.

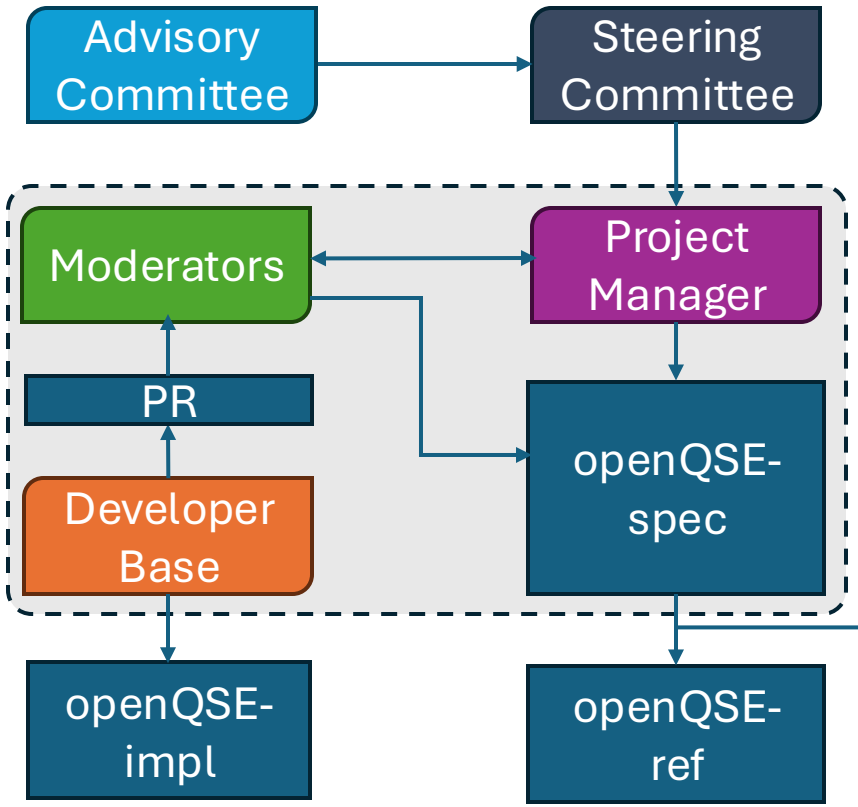
# openQSE Community Governance

Advisory Committee provides input on scope and community governance

Moderators administer openQSE development, e.g., pull requests

Developer base creates openQSE specification and implementations

openQSE-impl are potential independent implementations of the software specification




Steering Committee oversees governance of openQSE community and performs architectural reviews for long-term impact

Project Manager oversees daily openQSE development

openQSE-spec is a public reference specification of the software ecosystem

openQSE-ref is a public reference implementation of the software specification

openQSE-test is a public test suite to confirm adherence to the specification

# openQSE Change Control Process

- Documentation
  - Identify and document proposed changes, e.g., PR
- Assessment
  - Assess impact of proposed change on spec, code
  - Recommend action
- Decision
  - Review recommendations and make a decision, e.g., by majority vote
- Implementation
  - If approved, implement change
- Appeal
  - Refer complex cases to Steering Committee for guidance

# openQSE Communities

**Advisory Committee**

QHPC stakeholders
Accepting applications

**Developer Base**

IBM
IonQ
Quantinuum
IQM
Qblox
TUM
LBNL
STFC
RIKEN
ORNL
QSC
Open Applications

**Steering Committee**

ORNL
TUM
RIKEN
LBNL
ANL

**Project Manager**

Amir Shehata
Thomas Naughton

**Moderators**

Accepting applications

# openQSE Meeting Schedule

- Online Meetings
  - Weekly developer meetings with alternating schedule to span time zones
  - Slack channels
- In-person Meetings
  - Summer Workshop at ORNL
  - IEEE Quantum Week meet up
  - Supercomputing meet up
  - ISC meet up?